

Tutoring Module Business Requirements

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Revision History

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1.0	03/22/23	Initial document created.	John Leddo	John Leddo

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Overview

Purpose

The purpose of the Tutoring Module is to deliver personalized tutoring to each student based on his or her learning needs and learning style. The tutoring module represents an interaction of three important MEM system components: the learning management system which determines what needs to be taught and manages the interactions with the relevant databases, the learning environments that deliver the instruction and are the medium of student interaction with the system, and the AI engine which processes student actions in the learning environments and assesses what students have mastered and where their weaknesses are.

There are two modes of tutoring: student pick and computer pick. In the student pick mode, the student selects the topic(s) s/he wants tutoring on. The student receives instruction, practice and a post-test. In the computer pick mode, the topic is selected by the learning management system (LMS). Tutoring centers around the concept of instruction on a goal and/or learning objective. Therefore, there is a standard sequence for tutoring. The student is first given a pre-test for a specific goal. If the student performs below 90% learning objectives associated with the goal, s/he receives tutoring on those learning objectives in the order listed in the content database. Instruction consists of direct instruction, walkthrough problem solving, and practice problem solving. Tutoring and Assessments help take a student from their current level to their desired level of performance. A student's homework and assessments support their school assignments and aide in this effort.

The Learning Management System (LMS) collects data from each of these services in order to determine what shall be taught next in order to satisfy the program goals for the student.

Scope

The following components are within the scope of this release: This document is intended to cover all aspects of the tutoring service to be delivered under release 1 of MEM's online tutoring service.

Assumptions

The following assumptions apply to this project:

1. Any requirement not captured within this BRD shall be considered out of scope for this project and subsequent release.

Related Documents

The following reference materials are the authorities for issues not covered in this document:

1. Learning environments
 2. LMS
 3. Database
 4. AI engine
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Requirements

1. Managing the Tutoring Session

High-Level Description	The tutoring module will be accessed through the student portal. Once the student is in the tutoring module, s/he has the opportunity to decide whether s/he will pick the topic to be tutored or whether the system will pick the topic. Once the topic is selected, tutoring begins.			
Desired Behavior		Acceptance Criteria	Actual Test Results	
Req. #	Description	This requirement will be met if:	P / F	Results
1	The tutoring module shall be accessed either through the student portal after the student logs in.	The student can only select the tutoring module once logged in to the student portal.		
2	There shall be two modes of selecting a tutoring topic: student picks the topic and computer picks the topic. These are described in sections 2 and 3, respectively.	There are two modes of selecting the topic.		
3	When the computer picks the tutoring topics, tutoring shall proceed sequentially in order of the listed topics, then goals within each topic and then learning objectives for each topic.	Tutoring proceeds in order of the listed topics on the curriculum.		
4	A tutoring goal shall be considered mastered when the student records shows the student's performance to be at or above 90%.	The goal is mastered if performance is at or above the program goals.		
5	A tutoring topic/goal may be repeated or skipped depending upon whether the student records shows that the student has not met or met the program goals, respectively.	The tutoring topic/goal may be repeated or skipped depending upon whether program goals are met or not met.		
6	A typical tutoring sequence for a goal shall have the following default sequence: pretest, direct instruction for each learning objective, walkthrough problem solving for each learning objective, practice problems for each learning objective, and a post-test. If a student scores 90% or above on the pre-test, s/he may elect to skip that topic. If the student selects a topic for tutoring (student pick), s/he will not receive the pre-test. A session shall be considered "finished" upon successful completion of a post-test in which the student has met the program goals.	The tutoring sequence has the required components.		
7	If a student has started tutoring for a learning objective, but for some reason must pause or halt tutoring, the LMS must identify the most recently discussed material and pick up at that point when the student returns to complete tutoring at a later time, only if the student records still shows that learning objective to have not yet been mastered per the program goals. If the learning objective has been met, the student proceeds to the next learning objective for which program goals have not			

	been met.			
Other Considerations (if necessary)				
	None.			
Screen Captures (if necessary)				

2. Student Picks the Topic to be Tutored

High-Level Description	The MEM service will offer the student the option to select the topic, s/he wants to be tutored in or have the software pick. If the user opts to select the topic, s/he will be able to skip to the practice problem section of the lesson.			
	Desired Behavior	Acceptance Criteria	Actual Test Results	
Req. #	Description	This requirement will be met if:	P / F	Results
1	If the student selects the "student pick" option for tutoring, a list of available topics shall be presented. These will be color-coded to indicate the student's previous experience with them. Topics/learning objectives for which students have previously scored 90% or higher be color-coded green. Topics for which students have scored below the 90% level will be color-coded red. Topics for which there are no data (e.g., the student never took the initial diagnostic test and did not receive instruction on) will be color coded blue.	The available topics appear with appropriate color code.		
2	Once the student selects the topic, s/he will be able to skip to practice problems and skip the general instruction.	The student is given the choice of tutoring or practice.		
3	If the student asks for practice problems, s/he will be taken to the practice problem screen and given a minimum of 20 problems to practice. All practice problem functionality will be available. The student will continue to receive practice problems beyond the 20, if s/he has not met the 90% level after 20 problems. If s/he has, s/he will be asked whether s/he would like more problems or go on to something else.	The students will be given the 20 practice problems (or more if program goals not met) and then asked whether she'd like more problems or to go onto something else.		
4	If the student asks for more problems, then these will be added in groups of 10 as long as the student continues to request them. If the student's performance on the practice problems drops below 90%, s/he will continue to receive problems until the performance level reaches the program goals and the group of 10 has been done.	Additional problems are given as requested and will continue to be given if the student's level falls below the program goals.		
7	If the student indicates she would like to go onto something else, s/he will be given the option to pick another topic, have the computer pick or return to the general menu.	The choices will be available to the student.		
8	If the student indicates that she would like to practice, then the system will go to step 3 above.	The system responds appropriately.		
9	Upon completion of tutoring, the student will be given a post-test.	The post-test is given		
Other Considerations (if necessary)				

	None.
Screen Captures (if necessary)	

3. Computer picks topic to be tutored

High-Level Description	There shall be two modes of topic selection: student pick and computer pick. If the student elects to pick a topic, the LMS will check the student's records to find any session left off before completion, or lacking that, the first topic for which the student has not met the program goals. Once the topic is selected, the standard tutoring sequence applies.			
Desired Behavior		Acceptance Criteria	Actual Test Results	
Req. #	Description	This requirement will be met if:	P / F	Results
1	If the student indicates that s/he wants the computer to select a tutoring topic, then the learning management system will select a topic/goal for the student.	The LMS selects the topic for the student.		
2.	The LMS first checks to see if there were any sessions previously unfinished, where a student was working on a learning objective and had not yet met the program goals (90% performance level). If there is, then the LMS checks the student records to see if the program goals for this topic have been met (e.g., through a monthly assessment), if not, then the system resumes the previously unfinished session.	If there is an unfinished session and the program goals have not been met, then this session is continued.		
3	If there is no unfinished session or the student has met the program goals for the unfinished learning objective, then the LMS selects the first topic in order for which the program goals are not met.	The first topic for which program goals are not met is selected.		
4	If the student doesn't want to study the selected topic, s/he can press a "pick another topic" button and another topic will be selected in accordance with step 3.	The student can ask for another topic to be selected.		
5	The LMS then presents a pretest for the selected topic/goal.	A pretest is presented.		
6	If the student scores at or above the program goal on the pretest for that topic/goal, then the LMS returns to step 3. If the student does not score at or above the program goal for that topic/goal, then the LMS proceeds to direct instruction.	The student either gets a new topic or proceeds to direct instruction, depending upon whether or not s/he meets program goals on the pretest.		
Other Considerations (if necessary)				
	None.			
Screen Captures (if necessary)				

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4. Direct Instruction

High-Level Description		The instructional components of the tutoring session begins with direct instruction that presents the concept to the student. This will typically have the learning objective in the top part of the screen, a graphic on the left part of the screen and the explanation of the concept on the right part of the screen. The direct instruction will be optimized to match the student's known learning style and use the most effective prior content. The student can also request to be taught in a different way. When the student is ready, s/he is taken to walkthrough problem solving as the next step.		
Desired Behavior		Acceptance Criteria	Actual Test Results	
Req. #	Description	This requirement will be met if:	P / F	Results
1	The system will select direct instruction content from the database based on the lesson selected by the learning management system (LMS).	The appropriate content is selected.		
2	The content will be presented on the Direct Instruction screen. The learning objective will appear on the top of the screen. Any graphic will be displayed on the left side of the screen. The text will appear on the right side of the screen. In the event that the direct instruction has associated audio, it will be played while the direct instruction is on and cease to play if the student exits the screen. The student will have the option to adjust the volume of the audio. The student will have the option to replay the audio.	Each piece of content appears in the appropriate location on the screen. The audio is presented when appropriate. The audio is adjustable in volume. The audio disappears when the student leaves the screen.		
3	If the content on the right side of the screen is longer than can be contained in the field, there will be a scroll bar to allow the student to read the remaining text.	The field will scroll if the content is longer than the allowable screen space.		
4	Upon completion of the direct instruction, the LMS will record in the database that the students has taken the lesson.	The lesson is recorded in the database.		
Other Considerations (if necessary)				
	None.			
Screen Captures (if necessary)				

5. Walkthrough Problem Solving

High-Level Description	After a student has received direct instruction, s/he is shown a walkthrough problem to s/he can see how to solve problems of the type shown in the direct instruction. The walkthrough problem has to be learning objective specific since there may be different types of problems for each general goal. For example, the Pythagorean Theorem has several types of problems. So each one would be a separate learning objective with a separate direct instruction and walkthrough problem set. The walkthrough problem screen uses the same format as the direct instruction screen. At the top of the screen is the problem the student is learning to solve. A graphic of the problem will be on the left part of the screen and the explanation of how to solve the problem will be on the right part of the screen. When the student is ready, s/he is taken to practice problems as the next step.			
Desired Behavior		Acceptance Criteria	Actual Test Results	
Req. #	Description	This requirement will be met if:	P / F	Results
1	The system will select walkthrough problem solving content from the database.	The appropriate content is selected.		

2	The content will be presented on the Walkthrough Problem Solving screen. The problem description will appear on the top of the screen. Any graphic will be displayed on the left side of the screen. The text will appear on the right side of the screen. In the event that the walkthrough problem solving has associated audio, it will be played while the direct instruction is on and cease to play if the student exits the screen. The student will have the option to adjust the volume of the audio. The student will have the option to replay the audio.	Each piece of content appears in the appropriate location on the screen. The audio is presented when appropriate. The audio is adjustable in volume. The audio disappears when the student leaves the screen.		
3	If the content on the right side of the screen is longer than can be contained in the field, there will be a scroll bar to allow the student to read the remaining text.	The field will scroll if the content is longer than the allowable screen space.		
4	The system will record that the student completed the walkthrough section of the database.	The appropriate information is passed to the content database.		
Other Considerations (if necessary)				
	None.			
Screen Captures (if necessary)				

6. Practice Problems

High-Level Description	The system will present the student practice problems to ascertain the student's grasp of the content. If the student abandons a problem before correctly completing it by skipping to next problem, it is counted wrong. If there was work done and there was a mistake, the receives feedback. The student can also request hints at any time while solving the practice problems.			
Desired Behavior		Acceptance Criteria	Actual Test Results	
Req. #	Description	This requirement will be met if:	P / F	Results
1	The system will present no fewer than 20 problems to the student to determine if the student is performing at the level indicated by the student's program goals. Problems previously seen by the student will not be repeated.	No fewer than 20 practice problems are presented.		
2	If at the end of 20 practice problems, the student's performance on those practice problems has not met the program goals, then the student will continue to receive practice problems until the program goals are met.	The student continues to receive practice problems until the program goals are met.		
3	If at the end of 20 practice problems, the student's performance on those practice problems has met the program goals, then the student will be given the option to continue practice or exit practice problems.	The student is given the option to exit practice problems or continue.		
4	If the student opts to continue practice, s/he will be given problems in blocks of 10. If at the end	The student is given extra problems as described.		

	of these blocks, overall practice problem performance is not at the program goals level, s/he will receive more problems as in step 2. If the program goals are met, then the student will be given the choice to receive more or exit as in step 3.			
5	Upon completion of a problem, the student will go to the next problem by clicking a "next problem" button.	The student goes to the next problem by clicking a next problem button.		
6	In the event that the student clicks the next problem button before completing the current problem, a warning message will appear that the student failed to complete the current problem and then the student will be given the choice to go ahead anyway or stay with the same problem. If the student chooses to finish the problem, this will have the same effect as if the "next problem" button was not pressed at all.	The student is given the choice to go ahead or finish the current problem.		
7	If the student opts to go to the next problem, s/he will be given the option to toggle back and work on that problem again. If the student fails to correctly complete a practice problem, it will be counted as a failed problem from the point of view of meeting the program goals.	The student can return to a skipped problem but if s/he fails to successfully complete it, it will be counted as an incorrect problem.		
8	The practice problem screen shall have a description of the problem at the top of the screen. There will also be a toolbar at the top of the screen. There will be a worksheet and other learning environments that appear as needed on the screen. When a worksheet and another learning environment appear simultaneously, the worksheet will be on the right side of the screen and the other learning environment will be on the left. A feedback area will be on the bottom of the screen for presentation of any system-generated messages. There will be a hint button and a "done" button to indicate that the student has completed the problem.	The practice problem screen will contain each of the described components in the appropriate location.		
9	The toolbar will contain icons representing toolbar items that the student may need to solve problems. When clicked, a toolbar item will appear on the screen and remain until closed. The functionality for each toolbar item, including whether student actions while using that item are recorded for the AI engine will be described in a separate document.	The toolbar acts as described.		
10	The worksheet will resemble a sheet of lined writing paper. The student will use it to show work, one step per line. The student can change his work any time on any line by replacing old text with new. Any actions performed on the worksheet will be recorded and passed to the AI engine when the student presses the "done" button indicating completion of the current step.	The worksheet acts as described.		

11	Other learning environments will perform according to their own requirements documents. Actions performed on these learning environments will be recorded and passes to the AI engine when the student presses the “done” button.	The other learning environments perform according to their own requirements.		
12	When the student presses the hint button, the system will respond according to the hints section of this document.	The system responds according to the hints section.		
13	When the student presses, the “done” button, the system sends the student’s work and associated problem information to the AI engine for processing. The manner of this communication is described in a separate document.	The student’s work is passed to the AI engine.		
14	The AI engine will process the student’s work to detect mistakes.	The AI engine searches for the source of the error.		
15	If there is no mistake, then the problem will be counted as correct and the next line in the worksheet will become available for students to continue their work.	The next line becomes available.		
16	Upon the detected mistake, the system will search the AI database for a matching error. If a matching error is detected, the corresponding feedback will be presented. The problem will be marked as incorrect for purposes of meeting the 90% level.	The system checks the AI database for the source of the error and the problem is marked as incorrect.		
17	If a matching error cannot be found, the system will present the default feedback associated with the step in which the error occurred.	Default feedback is presented if an error match is not found.		
18	Once the source of the error has been detected, the system will present feedback in accordance with the requirements described in the feedback section of this document.	Feedback is presented once the source of the error is detected.		
19	Once the feedback is presented, the student will be given the opportunity to correct the problem. This will be done by returning to the worksheet or other learning environment and redoing the faulty work. The student will indicate s/he is done correcting the work by pressing the “done” button again. This repeats the above process, starting at step 13, although the student will not have the problem scored an additional time as correct or incorrect for purposes of meeting the program goals.	The student can correct his/her work but it is not re-counted toward the program goals.		
20	If the student does not know how to correct the error, s/he can press the hint button as described later in this document.	The hint button is available for the student and gives the appropriate hints.		
21	As needed, some problems will require students to answer the questions by selecting from multiple choice answers. With such questions, if the student presses the “done” button without answering the question by selecting one of the multiple-choice options, a message will popup notifying the student that s/he has not selected an answer. The AI engine will process the student’s step-by-step	Multiple choice answers are supported.		

	work leading to the answer just as with ordinary problems. However, if the student enters the answer on the worksheet and not in the multiple choice space, the answer will be disregarded.			
22	Practice problems will be able to support multiple required answers. When this happens, fields labeled for each required answer will be presented to the student, so that the student will know where to place each answer. The required answers can be numerical or multiple choice or using a learning environment (such as graphing an equation). In the case, where the answer is to be entered in a learning environment, there will be no separate field for those answers. However, if the answers require multiple copies of the same learning environment (e.g., creating two pie charts), then two copies of the learning environment will appear on the same screen for the student.	Multiple answers are supported.		

Other Considerations (if necessary)				

Screen Captures (if necessary)				
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7. Hints

High-Level Description	The system will present, upon student request, up to three levels of hint for any step the student is stuck on. Subsequent requests for hints will be processed by providing the additional hints associated with that step.			
Desired Behavior		Acceptance Criteria	Actual Test Results	
Req. #	Description	This requirement will be met if:	P / F	Results
1	There will be three different hints for each step/node. If this is the first time the student has requested a hint for that node, the student will be given the first available hint. If this is the second time the student has requested a hint for that node, the student will be given the second available hint. If this is the third time the student has requested a hint for that node, the student will be given the third available hint. On the fourth and subsequent hint requests for a given node, the third hint will appear along with the message, "Try doing this and see what you get."	There will be three hints in the DB per node. These will be presented in sequence upon multiple hint requests for the step. On the fourth try, the third hint and the message appear.		
2	The first hint (called a goal hint) will be a general hint that describes what a student is trying to accomplish. For example, the general hint for the equation $x + 5 = 8$ might be, "You want to isolate the variable."	The first hint is general.		
3	The second hint (called a step hint) will describe the type of step needed to accomplish what is stated in the first hint. The second hint for the equation $x + 5 = 8$ might be "Take the additive inverse of the constant and apply it to both sides of the equation."	The second hint appears as described.		
4	The third hint (called a problem hint) for the equation $x + 5 = 8$ might be "Subtract 5 from both sides of the equation to get x". Note that this uses actual information from the problem.	The third hint is specific and uses information from the problem.		
5	If the student presses the hint button prior to completing any work on the practice problem, the hint associated with the first step will be presented.	The hint associated with the first step is presented if the student presses the hint button without doing any work.		
Other Considerations (if necessary)				
	None.			
Screen Captures (if necessary)				

8. Feedback

High-Level Description	When a student completes a problem step, the problem type, step number and the student input will be passed to the AI engine for evaluation. The AI engine then evaluates the student's work. If there are no errors, then the next line in the worksheet becomes available. If the step in question completes the problem (i.e., provides the correct answer), the student receives a well done message and the session continues. If an error is detected, then the AI engine matches the student input to the error list in the AI database. If a
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	match is made, then the system returns the feedback associated with that error. If no match is made, then the AI engine retrieves the default feedback associated with the AI node for which a mistake was made. This feedback is then presented to the student in the lower section of the practice problem screen and the step at which the error occurred is highlighted. The student is instructed to fix the error and continue the problem. If the student still needs more help, s/he can press a button and see a similar problem with a similar mistake and the solution shown for that problem.			
Desired Behavior		Acceptance Criteria	Actual Test Results	
Req. #	Description	This requirement will be met if:	P / F	Results
1	When the student presses the done button on the practice problem screen, a description of the problem and the student’s work is passed to the AI engine for evaluation. If the student has done no work, a message will appear saying, “You haven’t done any work on the problem yet. If you are stuck, please press the hint button.”	The correct data are passed to the AI engine or the message appears if no work is done.		
2	The AI engine evaluates the students work to detect if the students work matches a correct step or an error. Once an error has been made, the AI engine will identify what specific error was made.	The AI engine identifies the AI nodes associated with an error and matches the student work to a specific error in the database.		
3	Once the error has been detected, the AI engine retrieves the feedback associated with that error. This feedback is presented in the bottom portion of the problem-solving screen. The step in which an error occurred shall be highlighted. The student shall be instructed to go back to the step in which the mistake was made and correct his or her work.	The AI engine retrieves the appropriate feedback and presents it along with the highlighted step and instructions to correct the student’s work.		
4	In the event the student’s work matches neither a correct solution nor a specific error, the AI engine will select the default feedback for that step to be presented.	The AI engine retrieves the default feedback and presents it along with the highlighted step and instructions to correct the student’s work.		
5	If the student cannot correct the mistake after receiving feedback, s/he may press the hint button as described previously in this document.	The student can press the hint button if s/he still cannot correct the problem and receive hints.		
Other Considerations (if necessary)				
	None.			
Screen Captures (if necessary)				

9. Student Queries

Desired Behavior		Acceptance Criteria	Actual Test Results	
Req. #	Description	This requirement will be met if:	P / F	Results
1	A query button will appear on the screen throughout all phase of the instructional session, including the homework help module..	The button appears.		

2	When the button is clicked, the user will have the opportunity to speak or type in a question.	Both input formats exist.		
3	When the user completes a verbal query, it will appear in the textbox as if the student spoke the query.	The query appears in the textbox as spoken.		
4	The user can edit the query.	The student can edit the query.		
5	When the user is satisfied that the query is correctly transcribed, s/he can press a submit button and the query will be submitted to the system.	Pressing the submit button transmits the query to the system.		
6	The system will match the user's query to the studentQA database. If a 90% match exists, the system will present the response associated with the query both in verbal and written formats.	The correct associated response is presented in both written and verbal formats if the match is greater than or equal to 90%.		
7	If the match in item 6 above is less than 90% but greater than 50%, the system will pick the highest matching query and present it to the user with the question, "Is this what you are asking?"	The highest match is presented.		
8	If the student responds "yes" to the query presented in item 7, the system will present the answer associated with that query as in item 6.	A yes response produces the associated answer to the query.		
9	If the student responds "no" to the query presented in item 7, then the system will present the next highest match and continue cycling until either the student answers "yes" or there are no further matches at the 50% level or above.	If the student responds no, the system cycles until the student responds "yes" or no more matches exist above 50%.		
10	If there are no (remaining) matches at 50% or above, the system will ask "can you rephrase the question"	The system asks the student to rephrase the question if there is no match at 50% or above.		
11	If the student submits a rephrased question, the system treats the rephrased question in accordance with item 6.	The system evaluates the rephrased question and looks for a 90% match.		
12	If the student cannot rephrase the question (indicates "no") or the rephrased question does not result in a match, the system responds that it cannot answer the question.	A failed effort to answer the question results in the system telling the student it cannot answer the question.		
13	Failed questions are stored in a database until the studentQA database is updated to accommodate the questions.	Failed questions are stored in a database.		
Other Considerations (if necessary)				
	None.			
Screen Captures (if necessary)				

10. System Queries

Desired Behavior		Acceptance Criteria	Actual Test Results	
Req.	Description	This requirement will be	P /	Results

#		met if:	F	
1	Upon completion of the direct instruction and walkthrough problems, the system will query the user to see if s/he understood the main points of the lesson. The user initiates this by pressing a button after the lesson. This is a skippable step.	The button appears.		
2	When the button is clicked, the user will have the system will speak and type questions for the user. Graphics may be used if necessary.	Both input formats exist.		
3	Queries will cover factual, procedural and conceptual knowledge covered in the lesson and will require answers that can be spoken, but not require the user to perform behavioral actions like graphing.	The appropriate queries are presented for each lesson.		
4.	When the user completes a verbal answer, it will appear in the textbox as if the student spoke the query.	The response appears in the textbox.		
5	The user can edit the response.	The student can edit the response.		
6	When the user is satisfied that the query is correctly transcribed, s/he can press a submit button and the response will be submitted to the system.	Pressing the submit button transmits the response to the system.		
6	The system will match the user's query to the systemQA database. If a 90% match exists, the system will evaluate whether the match represents a correct answer.	The system evaluates whether a correct answer was given if the match to an answer is 90%.		
7	If the match is to a correct answer, the system will respond with a positive response and proceed to the next question.	The system responds with a positive response and goes to the next question.		
8	If the match is to a negative answer or the match to any of the stored answers is less than 50%, the system responds by providing the correct answer.	The system provides the correct answer.		
9	If the match in item 6 above is less than 90% but greater than 75%, the system will pick the highest matching query and present it to the user with the question, "Is this what you are asking?"	The highest match is presented.		
10	If the student responds "yes" to the query presented in item 9, the system will evaluate the response as in item 7.	A yes response produces the associated answer to the response.		
11	If the student responds "no" to the query presented in item 7, then the system will present the next highest match and continue cycling until either the student answers "yes" or there are no further matches at the 75% level or above.	If the student responds no, the system cycles until the student responds "yes" or no more matches exist above 75%.		
12	If there are no (remaining) matches at 75% or above, the system will ask "can you rephrase the question"	The system asks the student to rephrase the question if there is no match at 75% or above.		
13	If the student submits a rephrased question, the system treats the rephrased question in accordance with item 7.	The system evaluates the rephrased question and looks for a 90% match.		

14	If the student cannot rephrase the question (indicates “no”) or the rephrased question does not result in at least a 75% match, the system provides the correct response.	A failed effort to answer the question results in the system telling the student the correct answer.		
15	Failed responses are stored in a database for use in reporting and for our own improvement of content.	Failed questions are stored in a database.		
Other Considerations (if necessary)				
	None.			
Screen Captures (if necessary)				

11. Remedial Instruction for learning objectives missed in the final assessment

High-Level Description		If the student fails to achieve a 90% performance level on a post-test for a given goal, s/he will receive remedial instruction on any learning objective for which the students missed problems on the post-test. The remedial instruction will be a scaled-down form of direction instruction, followed by a minimum of 10 practice problems. The student will continue working on practice problems until s/he meets the level set in the program goals, after which, s/he will move to the next learning objective for AI nodes are identified as not mastered. Once all of the learning objectives have been retaught, the student will take another post-test.		
Desired Behavior		Acceptance Criteria	Actual Test Results	
Req. #	Description	This requirement will be met if:	P / F	Results
1	If the student fails to score on a post-test for a goal at the level stated in the program goals, the student will receive remedial instruction. Remedial instruction will be given for any learning objective for which the student missed associated problems on the post-test.	The student receives remedial instruction for any learning objectives for problems missed on the goal post-test.		
2	Each learning objective requiring remediation will be remediated in the order in which they appear in the content database. For each learning objective to be remediated, remedial instruction shall consist of a scaled down form of direct instruction, followed by a minimum of 10 practice problems, followed by another post-test.	Remediation of learning objectives occurs in the appropriate sequence and has the required components.		
3	The AI engine shall start the remedial instruction by selecting remedial direct instruction from the content database. The selection of remedial direct instruction shall be done in the same manner as regular direct instruction and appear on the regular direct instruction screen.	The appropriate content is selected.		
4	The student will do a minimum of ten practice problems and continue to do them until s/he meets the performance level set in the program goals. Once this is met. The student takes another post-test.	The student is given practice problems until s/he meets the program goals.		
Other Considerations (if necessary)				
	None.			
Screen Captures (if necessary)				

12. Recording Session Activities

High-Level Description	Upon termination or completion of a session, the system shall record the following data in the database: details of the session, topics/learning objectives covered, content given to the student, activities performed, student performance. If a session was interrupted prior to completion, enough data will be saved such that the session can be reconstructed at the point of termination (except for where the student was in mid-problem solving).			
Desired Behavior		Acceptance Criteria	Actual Test Results	
Req. #	Description	This requirement will be met if:	P / F	Results
1	Upon termination or completion of a tutoring session, the system shall record, in the database, information about details of the session, topics/learning objectives covered, content given to the student, activities performed, student performance. If a session was interrupted prior to completion, enough data will be saved such that the session can be reconstructed at the point of termination (except for where the student was in mid-problem solving). These data shall be linked to the student who received the tutoring.	The data are recorded.		
2	Session details shall include the data and time of the tutoring, whether the tutoring was a scheduled tutoring date or optional, if scheduled, whether the session was completed according to its scheduled length, its length, and whether the computer or student selected the tutoring topics.	The appropriate session details are recorded.		
3	The system shall record the topics and learning objectives that were covered by the student.	The topics and learning objectives are recorded.		
4	The system will record all of the activities performed by the student by each learning objective, including direct instruction (how many versions were seen), walkthrough problem solving (how many versions were seen), practice problems (how many were given), and remedial instruction (how many versions were seen).	The activities are recorded.		
5	The system will record the student performance on the practice problems. This includes both the number correct and incorrect as well as updating the student's record in the AI database with any concepts the student has not mastered. A problem shall be considered correct if the student completed it without any errors. If a problem had one or more errors or was not completed, it shall be considered to be incorrect.	The performance data is recorded.		
6	If the student has met the program goals for a learning objective, that learning objective will be marked as completed. Otherwise, the session will be considered to be interrupted or unfinished. If a tutoring session is interrupted prior to completion, the details of the existing session (except for the student's work on a current practice problem) will be saved such that the session could be continued at a later	The learning objective is marked as completed if the program goals are met. The details of the existing session are saved if the session is interrupted or not completed.		

	date.			
Other Considerations (if necessary)				
	None.			
Screen Captures (if necessary)				

13. Approval

By signing below, the Business confirms the following:

1. The Business Requirements Document (BRD) has been provided and received by the Business.
2. The Acceptance Criteria outlined within the BRD in joint conversation between the Requirements Analyst and the Business accurately capture the Business' expectations for this project release and the criteria that the Business will use to validate this project release.

Business Manager 1	Business Manager 2
Signature	Signature
Printed Name	Printed Name
Title	Title
Date	Date

Appendices

14. If Needed